AMORO Lab: Kinematics and Dynamics of a Biglide

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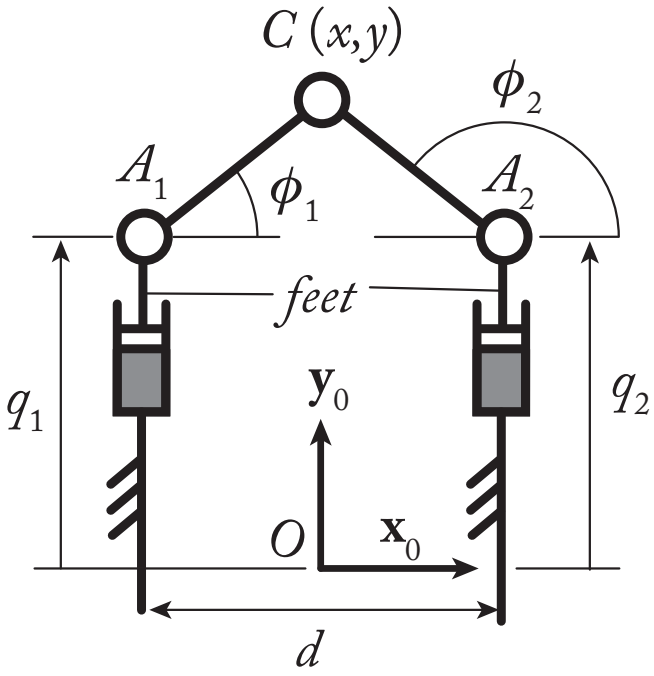


Figure 1: Kinematic model of the Biglide parallel mechanism. The two prismatic vertical sliders are actuated.

1 Model

The purpose of this laboratory work is to model, simulate, and control a 2-DOF planar parallel manipulator known as the **Biglide** mechanism. Parallel kinematic manipulators (PKMs) provide distinct advantages over serial robots in terms of high payload-to-weight ratios, structural stiffness, and high operational accelerations. However, their closed kinematic chain architecture results in complex nonlinear kinematic mappings, multiple assembly and working modes, coupled dynamics, and internal singular configurations.

The kinematic architecture of the Biglide mechanism is illustrated in Figure 1. The robot consists of two actuated prismatic joints (sliders or feet) moving along parallel vertical rails, and two passive rigid binary links (legs) connecting the feet to a common point $C(x, y)$, which serves as the end-effector.

For the Gazebo robotic platform considered in this work, the benchmark geometric and dynamic parameters are defined as:

- Rail separation distance: $d = 0.4$ m.
- Length of passive binary links: $l_{A_1C} = l_{A_2C} = l = 0.2828427$ m $\approx \frac{0.4}{\sqrt{2}}$ m.
- Mass of the moving platform / end-effector: $m_p = 3.0$ kg (this lumped mass incorporates the end-effector cylinder and the concentrated mass contributions of the two legs located at joint C).
- Mass of each actuated slider (foot): $m_f = 1.0$ kg.
- The mechanism operates in the horizontal XY-plane (perpendicular to the vertical gravity vector along $-Z$); consequently, the gravitational force projection along the active prismatic axes is strictly zero ($\mathbf{g} = \mathbf{0}$).

This document details the analytic derivation of the geometric, first-order kinematic, second-order kinematic, and dynamic models of the Biglide. These models are implemented in `student_scripts/biglide_models.py`. Their validation against the Gazebo simulator, the Computed Torque Controller and the Type 2 singularity study are reported in the companion document *Amoro_Lab_Report_ELNAGGAR_PANDIAN.pdf*.

2 Geometric Models

2.1 Direct Geometric Model (DGM)

The Direct Geometric Model (DGM) determines the Cartesian position of the end-effector $C(x, y)$ given the displacements of the active prismatic joints $\mathbf{q}_a = [q_1, q_2]^T$.

Referring to the base coordinate frame $\mathcal{F}_0(O, \mathbf{x}_0, \mathbf{y}_0)$ centered midway between the rails at $y = 0$, the coordinates of the actuated sliders A_1 and A_2 are:

$$A_1 = \begin{bmatrix} -d/2 \\ q_1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} d/2 \\ q_2 \end{bmatrix} \tag{1}$$

The kinematic loop-closure equations enforce the constant distance constraints of the rigid legs A_1C and A_2C:

$$\left(x + \frac{d}{2}\right)^2 + (y - q_1)^2 = l^2 \tag{2}$$

$$\left(x - \frac{d}{2}\right)^2 + (y - q_2)^2 = l^2 \tag{3}$$

Subtracting (3) from (2) eliminates the quadratic terms in x and y:

$$2dx - 2q_1y + q_1^2 + 2q_2y - q_2^2 = 0 \tag{4}$$

Rearranging gives an explicit linear relationship expressing x as a function of y:

$$x = \frac{q_1 - q_2}{d}\left(y - \frac{q_1 + q_2}{2}\right) = Ay + B \tag{5}$$

where the geometric coefficients A and B are defined by:

$$A = \frac{q_1 - q_2}{d}, \quad B = -A\left(\frac{q_1 + q_2}{2}\right) = -\frac{q_1^2 - q_2^2}{2d} \tag{6}$$

Substituting (5) back into the loop-closure equation (2) yields a quadratic scalar equation in y:

$$a_y y^2 + b_y y + c_y = 0 \tag{7}$$

where:

$$a_y = 1 + A^2 \tag{8}$$

$$b_y = 2\left[A\left(B + \frac{d}{2}\right) - q_1\right] \tag{9}$$

$$c_y = \left(B + \frac{d}{2}\right)^2 + q_1^2 - l^2 \tag{10}$$

The discriminant is given by $\Delta = b_y^2 - 4a_y c_y$. For physically realizable configurations within the operational workspace, $\Delta \geq 0$. The two mathematical roots correspond to the two distinct assembly modes of the Biglide:

$$y = \frac{-b_y - m_a\sqrt{\Delta}}{2a_y} \tag{11}$$

where $m_a \in \{-1, +1\}$ denotes the assembly mode parameter.

- $m_a = -1$ corresponds to the **top assembly mode** ($y > q_i$, where the platform is located above the sliders). In the Gazebo simulation and the lab evaluation scripts, the robot operates exclusively in this mode.
- $m_a = +1$ corresponds to the inverted bottom assembly mode ($y < q_i$).

Once y is computed, the horizontal position x is directly obtained from (5).

2.2 Passive Joints Geometric Model

The passive joint variables $\mathbf{q}_d = [\phi_1, \phi_2]^T$ represent the absolute orientation angles of links A_1C and A_2C with respect to the horizontal $\mathbf{x}_0$-axis. Because the active joints are pure prismatic sliders with fixed vertical axes, the relative angles measured by the encoders at A_1 and A_2 correspond identically to the absolute angles ϕ_1 and ϕ_2.

From the vector geometry $\overrightarrow{A_1C} = C - A_1$ and $\overrightarrow{A_2C} = C - A_2$:

$$\phi_1 = \text{atan2}\left(y - q_1, x + \frac{d}{2}\right) \tag{12}$$

$$\phi_2 = \text{atan2}\left(y - q_2, x - \frac{d}{2}\right) \tag{13}$$

For the nominal symmetric reference pose $\mathbf{q}_a = [0, 0]^T$, the platform is located at $x = 0$, $y = \sqrt{l^2 - (d/2)^2} = \sqrt{0.2828427^2 - 0.2^2} = 0.20$ m. In this configuration, $\phi_1 = \text{atan2}(0.2, 0.2) = \frac{\pi}{4}$ rad ≈ 0.7854 rad, and $\phi_2 = \text{atan2}(0.2, -0.2) = \frac{3\pi}{4}$ rad ≈ 2.3562 rad, matching the Gazebo initial state.

2.3 Inverse Geometric Model (IGM)

The Inverse Geometric Model (IGM) determines the active joint positions q_1, q_2 required to achieve a prescribed end-effector position $C(x, y)$.

Rearranging the loop-closure equations (2) and (3):

$$(y - q_1)^2 = l^2 - \left(x + \frac{d}{2}\right)^2 \tag{14}$$

$$(y - q_2)^2 = l^2 - \left(x - \frac{d}{2}\right)^2 \tag{15}$$

Taking the square root yields:

$$q_1 = y + \mu_1 \sqrt{l^2 - \left(x + \frac{d}{2}\right)^2} \tag{16}$$

$$q_2 = y + \mu_2 \sqrt{l^2 - \left(x - \frac{d}{2}\right)^2} \tag{17}$$

where $\mu_1, \mu_2 \in \{-1, +1\}$ represent the working modes of Leg 1 and Leg 2. For the Biglide configuration where the moving platform is held above the actuated feet, the sliders must satisfy $q_i < y$, which corresponds to $\mu_1 = -1$ and $\mu_2 = -1$.

3 First-Order Kinematic Models

3.1 Differential Loop-Closure and Constraint Equations

Let $\mathbf{u}_1$ and $\mathbf{u}_2$ denote the unit vectors directed along the passive links A_1C and A_2C, and let $\mathbf{v}_1, \mathbf{v}_2$ be the unit vectors obtained by rotating $\mathbf{u}_1, \mathbf{u}_2$ by $+90°$ counter-clockwise:

$$\mathbf{u}_i = \begin{bmatrix} \cos\phi_i \\ \sin\phi_i \end{bmatrix}, \quad \mathbf{v}_i = \begin{bmatrix} -\sin\phi_i \\ \cos\phi_i \end{bmatrix}, \quad i \in \{1, 2\} \tag{18}$$

The position of point C expressed along each leg is:

$$\mathbf{r}_{OC} = \begin{bmatrix} -d/2 \\ q_1 \end{bmatrix} + l\mathbf{u}_1 \tag{19}$$

$$\mathbf{r}_{OC} = \begin{bmatrix} d/2 \\ q_2 \end{bmatrix} + l\mathbf{u}_2 \tag{20}$$

Differentiating with respect to time t, noting that $\dot{\mathbf{u}}_i = \dot{\phi}_i \mathbf{v}_i$:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 \\ \dot{q}_1 \end{bmatrix} + l\dot{\phi}_1 \mathbf{v}_1 = \dot{q}_1 \mathbf{j}_0 + l\dot{\phi}_1 \mathbf{v}_1 \tag{21}$$

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 \\ \dot{q}_2 \end{bmatrix} + l\dot{\phi}_2 \mathbf{v}_2 = \dot{q}_2 \mathbf{j}_0 + l\dot{\phi}_2 \mathbf{v}_2 \tag{22}$$

where $\dot{\mathbf{x}} = [\dot{x}, \dot{y}]^T$ and $\mathbf{j}_0 = [0, 1]^T$.

To eliminate the unknown passive joint velocities $\dot{\phi}_1$ and $\dot{\phi}_2$, we multiply each vector equation by $\mathbf{u}_i^T$. Since $\mathbf{u}_i \cdot \mathbf{v}_i = 0$ and $\mathbf{u}_i \cdot \mathbf{j}_0 = \sin\phi_i$:

$$\mathbf{u}_1^T \dot{\mathbf{x}} = \dot{q}_1 \sin\phi_1 \implies \cos\phi_1 \dot{x} + \sin\phi_1 \dot{y} = \sin\phi_1 \dot{q}_1 \tag{23}$$

$$\mathbf{u}_2^T \dot{\mathbf{x}} = \dot{q}_2 \sin\phi_2 \implies \cos\phi_2 \dot{x} + \sin\phi_2 \dot{y} = \sin\phi_2 \dot{q}_2 \tag{24}$$

Assembling into standard matrix form:

$$\mathbf{A}_r \dot{\mathbf{x}} + \mathbf{B}\dot{\mathbf{q}}_a = \mathbf{0} \tag{25}$$

where $\dot{\mathbf{q}}_a = [\dot{q}_1, \dot{q}_2]^T$, and the constraint matrices are:

$$\mathbf{A}_r = \begin{bmatrix} \cos\phi_1 & \sin\phi_1 \\ \cos\phi_2 & \sin\phi_2 \end{bmatrix}, \quad \mathbf{B} = -\begin{bmatrix} \sin\phi_1 & 0 \\ 0 & \sin\phi_2 \end{bmatrix} \tag{26}$$

3.2 Inverse Kinematic Model (IKM)

Assuming $\mathbf{B}$ is invertible ($\sin\phi_1 \neq 0, \sin\phi_2 \neq 0$):

$$\dot{\mathbf{q}}_a = -\mathbf{B}^{-1}\mathbf{A}_r\dot{\mathbf{x}} = \mathbf{J}^{-1}\dot{\mathbf{x}} \tag{27}$$

where the inverse Jacobian matrix is:

$$\mathbf{J}^{-1} = \begin{bmatrix} \frac{1}{\sin\phi_1} & 0 \\ 0 & \frac{1}{\sin\phi_2} \end{bmatrix} \begin{bmatrix} \cos\phi_1 & \sin\phi_1 \\ \cos\phi_2 & \sin\phi_2 \end{bmatrix} = \begin{bmatrix} \cot\phi_1 & 1 \\ \cot\phi_2 & 1 \end{bmatrix} \tag{28}$$

In scalar form:

$$\dot{q}_1 = \cot\phi_1\dot{x} + \dot{y} \tag{29}$$

$$\dot{q}_2 = \cot\phi_2\dot{x} + \dot{y} \tag{30}$$

3.3 Forward Kinematic Model (FKM)

Assuming $\mathbf{A}_r$ is non-singular ($\det(\mathbf{A}_r) \neq 0$):

$$\dot{\mathbf{x}} = -\mathbf{A}_r^{-1}\mathbf{B}\dot{\mathbf{q}}_a = \mathbf{J}\dot{\mathbf{q}}_a \tag{31}$$

where the direct Jacobian matrix $\mathbf{J}$ is the inverse of $\mathbf{J}^{-1}$:

$$\mathbf{J} = (\mathbf{J}^{-1})^{-1} = \frac{1}{\cot\phi_1 - \cot\phi_2} \begin{bmatrix} 1 & -1 \\ -\cot\phi_2 & \cot\phi_1 \end{bmatrix} \tag{32}$$

In scalar form:

$$\dot{x} = \frac{\dot{q}_1 - \dot{q}_2}{\cot\phi_1 - \cot\phi_2} \tag{33}$$

$$\dot{y} = \frac{-\cot\phi_2\dot{q}_1 + \cot\phi_1\dot{q}_2}{\cot\phi_1 - \cot\phi_2} \tag{34}$$

3.4 Passive Joint Velocities Model

Multiplying the velocity equations (21) and (22) by $\mathbf{v}_i^T$ (since $\mathbf{v}_i \cdot \mathbf{v}_i = 1$ and $\mathbf{v}_i \cdot \mathbf{j}_0 = \cos\phi_i$):

$$\dot{\phi}_1 = \frac{-\sin\phi_1\dot{x} + \cos\phi_1(\dot{y} - \dot{q}_1)}{l} \tag{35}$$

$$\dot{\phi}_2 = \frac{-\sin\phi_2\dot{x} + \cos\phi_2(\dot{y} - \dot{q}_2)}{l} \tag{36}$$

In matrix form:

$$\dot{\mathbf{q}}_d = \mathbf{J}_{td}^{-1}\left(\mathbf{J}_t\dot{\mathbf{x}} - \mathbf{J}_{ta}\dot{\mathbf{q}}_a\right) \tag{37}$$

with:

$$\mathbf{J}_{td} = \begin{bmatrix} l & 0 \\ 0 & l \end{bmatrix}, \quad \mathbf{J}_t = \begin{bmatrix} -\sin\phi_1 & \cos\phi_1 \\ -\sin\phi_2 & \cos\phi_2 \end{bmatrix}, \quad \mathbf{J}_{ta} = \begin{bmatrix} \cos\phi_1 & 0 \\ 0 & \cos\phi_2 \end{bmatrix} \tag{38}$$

3.5 Singularity Analysis

The parallel Biglide mechanism exhibits three classes of singular configurations:

1. **Type 1 Singularity ($\det(\mathbf{B}) = 0$):**

$$\det(\mathbf{B}) = \sin\phi_1\sin\phi_2 = 0 \implies \phi_1 \in \{0, \pi\} \quad \text{or} \quad \phi_2 \in \{0, \pi\} \tag{39}$$

Physical Meaning: Occurs when either link A_1C or A_2C becomes completely horizontal ($y = q_1$ or $y = q_2$). At this boundary of the reachable workspace, vertical motion cannot be actuated.

2. **Type 2 Singularity ($\det(\mathbf{A}_r) = 0$):**

$$\det(\mathbf{A}_r) = \cos\phi_1 \sin\phi_2 - \sin\phi_1 \cos\phi_2 = \sin(\phi_2 - \phi_1) = 0 \implies \phi_2 - \phi_1 \in \{0, \pm\pi\} \quad (40)$$

Physical Meaning: Occurs when links A_1C and A_2C become parallel or collinear. The platform gains an unconstrained internal degree of freedom (infinitesimal mobility) while the actuators are locked.

3. **LPJTS Singularity ($\det(\mathbf{J}_{td}) = 0$):** $\det(\mathbf{J}_{td}) = l^2 \neq 0$ since $l > 0$. Therefore, this singularity never occurs.

4 Second-Order Kinematic Models

4.1 Differentiated Constraint Equations and Convective Accelerations

Differentiating the first-order constraint equation $\mathbf{A}_r\dot{\mathbf{x}} + \mathbf{B}\dot{\mathbf{q}}_a = \mathbf{0}$ with respect to time:

$$\mathbf{A}_r\ddot{\mathbf{x}} + \mathbf{B}\ddot{\mathbf{q}}_a + \mathbf{b}_p = \mathbf{0} \quad (41)$$

where $\mathbf{b}_p = \dot{\mathbf{A}}_r\dot{\mathbf{x}} + \dot{\mathbf{B}}\dot{\mathbf{q}}_a$ represents the vector of convective accelerations. Evaluating:

$$\dot{\mathbf{A}}_r = \begin{bmatrix} -\dot{\phi}_1 \sin\phi_1 & \dot{\phi}_1 \cos\phi_1 \\ -\dot{\phi}_2 \sin\phi_2 & \dot{\phi}_2 \cos\phi_2 \end{bmatrix}, \quad \dot{\mathbf{B}} = -\begin{bmatrix} \dot{\phi}_1 \cos\phi_1 & 0 \\ 0 & \dot{\phi}_2 \cos\phi_2 \end{bmatrix} \quad (42)$$

Evaluating the first component b_{p1}:

$$b_{p1} = \dot{\phi}_1 \left[-\sin\phi_1 \dot{x} + \cos\phi_1 (\dot{y} - \dot{q}_1) \right] \quad (43)$$

Using (35), the bracketed term is identically $l\dot{\phi}_1$. Therefore:

$$\mathbf{b}_p = \begin{bmatrix} l\dot{\phi}_1^2 \\ l\dot{\phi}_2^2 \end{bmatrix} \quad (44)$$

4.2 Inverse Second-Order Kinematic Model (IKM2)

From (41), assuming $\mathbf{B}$ is invertible:

$$\ddot{\mathbf{q}}_a = -\mathbf{B}^{-1}\mathbf{A}_r\ddot{\mathbf{x}} - \mathbf{B}^{-1}\mathbf{b}_p = \mathbf{J}^{-1}\ddot{\mathbf{x}} + \mathbf{a}_q \quad (45)$$

where the active convective acceleration $\mathbf{a}_q$ is:

$$\mathbf{a}_q = -\mathbf{B}^{-1}\mathbf{b}_p = \begin{bmatrix} \frac{l\dot{\phi}_1^2}{\sin\phi_1} \\ \frac{l\dot{\phi}_2^2}{\sin\phi_2} \end{bmatrix} \quad (46)$$

In scalar form:

$$\ddot{q}_1 = \cot\phi_1 \ddot{x} + \ddot{y} + \frac{l\dot{\phi}_1^2}{\sin\phi_1} \quad (47)$$

$$\ddot{q}_2 = \cot\phi_2 \ddot{x} + \ddot{y} + \frac{l\dot{\phi}_2^2}{\sin\phi_2} \quad (48)$$

4.3 Direct Second-Order Kinematic Model (DKM2)

From (41), assuming $\mathbf{A}_r$ is invertible:

$$\ddot{\mathbf{x}} = -\mathbf{A}_r^{-1}\mathbf{B}\ddot{\mathbf{q}}_a - \mathbf{A}_r^{-1}\mathbf{b}_p = \mathbf{J}\ddot{\mathbf{q}}_a + \mathbf{a}_t \quad (49)$$

where $\mathbf{a}_t = -\mathbf{A}_r^{-1}\mathbf{b}_p$ is the Cartesian convective acceleration:

$$\mathbf{a}_t = \frac{1}{\sin(\phi_2 - \phi_1)} \begin{bmatrix} -l\dot{\phi}_1^2 \sin\phi_2 + l\dot{\phi}_2^2 \sin\phi_1 \\ l\dot{\phi}_1^2 \cos\phi_2 - l\dot{\phi}_2^2 \cos\phi_1 \end{bmatrix} \quad (50)$$

4.4 Passive Joint Accelerations Model

Differentiating the velocity equations (21) and (22) with respect to time:

$$\ddot{\mathbf{x}} = \ddot{q}_i \mathbf{j}_0 + l\ddot{\phi}_i \mathbf{v}_i - l\dot{\phi}_i^2 \mathbf{u}_i, \quad i \in \{1, 2\} \tag{51}$$

Multiplying by $\mathbf{v}_i^T$ (noting that $\mathbf{v}_i \cdot \mathbf{v}_i = 1$ and $\mathbf{v}_i \cdot \mathbf{u}_i = 0$):

$$\ddot{\phi}_1 = \frac{-\sin\phi_1 \ddot{x} + \cos\phi_1(\ddot{y} - \ddot{q}_1)}{l} \tag{52}$$

$$\ddot{\phi}_2 = \frac{-\sin\phi_2 \ddot{x} + \cos\phi_2(\ddot{y} - \ddot{q}_2)}{l} \tag{53}$$

5 Inverse Dynamic Model

5.1 Virtual Tree Decomposition and Principle of Virtual Powers

Following the standard Lagrangian dynamic methodology for parallel manipulators, the closed kinematic chains are virtually cut at the passive joint C, forming a tree structure composed of:

1. Two actuated sliders moving along vertical rails: total kinetic energy $T_f = \frac{1}{2} m_f (\dot{q}_1^2 + \dot{q}_2^2)$.
2. The free moving platform concentrated at point C: kinetic energy $T_p = \frac{1}{2} m_p (\dot{x}^2 + \dot{y}^2)$.

In the Gazebo benchmark model, the masses of the two passive legs are concentrated at their extremities ($m_p = 3.0$ kg total lumped mass at C, $m_f = 1.0$ kg at each foot).

Applying Lagrange's equations to the unconstrained virtual system:

$$\boldsymbol{\tau}_t = m_f \ddot{\mathbf{q}}_a \quad \text{(Slider dynamics)} \tag{54}$$

$$\mathbf{w}_p = m_p \ddot{\mathbf{x}} \quad \text{(Platform wrench)} \tag{55}$$

By D'Alembert's Principle of Virtual Powers, the virtual power developed by all internal constraint reaction forces vanishes. For any virtual velocity $\delta\dot{\mathbf{q}}_a$, the compatible virtual velocity of the platform is:

$$\delta\dot{\mathbf{x}} = \mathbf{J}\delta\dot{\mathbf{q}}_a \implies \delta\dot{\mathbf{x}}^T = \delta\dot{\mathbf{q}}_a^T \mathbf{J}^T \tag{56}$$

Equating the virtual power of the generalized actuator forces to the virtual power of the acceleration forces:

$$\delta\dot{\mathbf{q}}_a^T \boldsymbol{\tau} = \delta\dot{\mathbf{q}}_a^T \boldsymbol{\tau}_t + \delta\dot{\mathbf{x}}^T \mathbf{w}_p = \delta\dot{\mathbf{q}}_a^T \left(\boldsymbol{\tau}_t + \mathbf{J}^T \mathbf{w}_p\right) \tag{57}$$

Since this holds for any arbitrary virtual velocity $\delta\dot{\mathbf{q}}_a$, we obtain the exact dynamic projection:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_t + \mathbf{J}^T \mathbf{w}_p \tag{58}$$

Notice that the projection uses the transpose of the **direct** Jacobian $\mathbf{J}^T$, in strict agreement with duality in kinematics and statics.

5.2 Analytical Mass Matrix and Convective Terms

Substituting $\ddot{\mathbf{x}} = \mathbf{J}\ddot{\mathbf{q}}_a + \mathbf{a}_t$ into (58):

$$\boldsymbol{\tau} = m_f \ddot{\mathbf{q}}_a + \mathbf{J}^T m_p \left(\mathbf{J}\ddot{\mathbf{q}}_a + \mathbf{a}_t\right) = \left(m_f \mathbf{I}_{2\times 2} + m_p \mathbf{J}^T \mathbf{J}\right) \ddot{\mathbf{q}}_a + m_p \mathbf{J}^T \mathbf{a}_t \tag{59}$$

This puts the inverse dynamic model into the canonical form:

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q}_a)\ddot{\mathbf{q}}_a + \mathbf{c}(\mathbf{q}_a, \dot{\mathbf{q}}_a) \tag{60}$$

where:

$$\mathbf{M}(\mathbf{q}_a) = m_f \mathbf{I}_{2\times 2} + m_p \mathbf{J}^T \mathbf{J} \tag{61}$$

$$\mathbf{c}(\mathbf{q}_a, \dot{\mathbf{q}}_a) = m_p \mathbf{J}^T \mathbf{a}_t \tag{62}$$

Expanding $\mathbf{J}^T\mathbf{J}$ explicitly using $\mathbf{J}$ from (32):

$$\mathbf{J}^T\mathbf{J} = \frac{1}{(\cot\phi_1 - \cot\phi_2)^2} \begin{bmatrix} 1+\cot^2\phi_2 & -(1+\cot\phi_1\cot\phi_2) \\ -(1+\cot\phi_1\cot\phi_2) & 1+\cot^2\phi_1 \end{bmatrix} \tag{63}$$

Thus:

$$\mathbf{M}(\mathbf{q}_a) = \begin{bmatrix} m_f + \frac{m_p(1+\cot^2\phi_2)}{(\cot\phi_1-\cot\phi_2)^2} & -\frac{m_p(1+\cot\phi_1\cot\phi_2)}{(\cot\phi_1-\cot\phi_2)^2} \\ -\frac{m_p(1+\cot\phi_1\cot\phi_2)}{(\cot\phi_1-\cot\phi_2)^2} & m_f + \frac{m_p(1+\cot^2\phi_1)}{(\cot\phi_1-\cot\phi_2)^2} \end{bmatrix} \tag{64}$$

The mass matrix $\mathbf{M}(\mathbf{q}_a)$ is symmetric and strictly positive-definite across the operational workspace outside of Type 2 singularities.